



Research paper

Design and implementation of an open-Source IoT and blockchain-based peer-to-peer energy trading platform using ESP32-S2, Node-Red and, MQTT protocol

Mirza Jabbar Aziz Baig^{a,*}, M. Tariq Iqbal^a, Mohsin Jamil^a, Jahangir Khan^b

^a Department of Electrical and Computer Engineering, Memorial University of Newfoundland, St. John's-A1C5S7, NL, Canada

^b BC Hydro, Transmission Lines Engineering, 6911 Southpoint Drive, Burnaby-V3N4X8, B.C, Canada



ARTICLE INFO

Article history:

Received 10 July 2021

Received in revised form 6 August 2021

Accepted 28 August 2021

Available online 12 September 2021

Keywords:

Peer-to-Peer (P2P)

Ethereum blockchain

Message Queuing Telemetry Transport (MQTT)

Hypertext Transfer Protocol (http)

Internet of things (IoT)

ABSTRACT

An open-source P2P energy trading platform facilitates energy trading amongst the peers. The proposed system provides real time data acquisition, monitoring and control of self-generated energy at a remote location. The trading activities are done on a web interface that uses a private Ethereum blockchain. A smart contract is deployed on the Ethereum blockchain and the trading activities performed on the web interface are recorded on a tamper-proof blockchain network. An internet of things platform is used to monitor and control the self-generated energy. Energy data is collected and processed by means of ESP32-S2 microcontrollers using field instrumentation devices which are connected to the voltage source and load. An open-source decentralized Peer-to-Peer (P2P) energy trading system, designed on the blockchain and internet of things (IoT) architecture is proposed. The hardware setup includes a relay, a current sensor, a voltage sensor, a Wi-Fi router and ESP32-S2 microcontroller. For data transfer the Message Queuing Telemetry Transport (MQTT) protocol is used over a local network. ESP32-S2 is set up as MQTT client and Node-Red IoT server is used as MQTT broker. Hypertext Transfer Protocol (http) request method is implemented to connect the Node-Red server with the web interface developed using ReactJS library. The system design, implementation, testing, and results are presented in this paper.

© 2021 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

The proliferation of Distributed Energy Resources (DER) such as Photovoltaics (PV) systems, Battery Energy Storage Systems (BESS), Electric Vehicles (EV), Wind Energy Conversion System (WECS) along with advancements in information and communication systems technologies, has introduced a new electricity market (Alskaif et al., 2021). A significant demand for electric energy and the availability of low-cost renewable energy technologies has raised the number of prosumers in the electricity market. As a result, a considerable interest of selling or sharing electrical energy has been witnessed in the electricity market (Baig et al., 2020). Blockchain and internet of things has given a new shape to electricity market. Rather than rely on centralized authority, peers can now perform transactions on their own. Blockchain technology and the Internet of Things can boost the efficiency of peer-to-peer energy trading platforms. Now, consumers can buy energy on local energy markets, instead of having to buy it from utilities. Blockchain and internet of things-based

energy trading platforms can be more efficient and controllable for prosumers. The immutable blockchain ledger can also provide a transparent energy and funds usage data in real-time.

As the residential sector is increasingly adopting distributed energy resources, several new market approaches will be necessary. Market participants now prefer distributed generation and to sell electrical power, contrarily to the conventional electricity market. This emerging concept of prosumers now requires a state-of-the-art platform for energy trading. A platform that can provide full trading autonomy, the financial security and pricing liberty and to choose the best suitable time for trading sessions. The direct exchange of energy amongst consumers and prosumers is known as P2P energy trading. Over the past few years, peer-to-peer (P2P) trading has become an only option for consumers to participate actively in energy market. Peers can trade excess energy production with peers through peer-to-peer energy trading, which is beneficial for both parties (Soto et al., 2021). Consequently, their investment can yield a return. An important part of peer-to-peer energy trading is that it helps prosumers to trade their energy similar to goods and services. In addition, energy trading on a peer-to-peer basis can also help reduce peak demand on the grid, lowering the need for reserves,

* Corresponding author.

E-mail address: mjabaig@mun.ca (M.J.A. Baig).

and reducing network outages (Tushar et al., 2020a). As described by the GridWise Architecture Council, P2P is also defined as trans-active energy (TE), a system for dynamically managing supply and demand across the electric grid. The P2P platform industry is expected to generate revenues of over \$4 billion by 2026 (AlAshery et al., 2021).

A blockchain-based P2P energy trading system is presented in AlAshery et al. (2021). The system includes bilateral contracts, the Vickrey-Clarke-Groves (VCG) mechanism, an electronic commerce platform, and the ability to trade with the main grid. This multi-layered mechanism accommodates changes in electricity generation and consumption preferences. Furthermore, the VCG mechanism minimizes market power by ensuring that bids are truthful and transparent. To compensate for the disadvantage of VCG, namely, the unbalanced budget, different remedies are also proposed. In their description of the trading framework, the authors described it as multi-settlement and quasi-ideal. To evaluate the execution of proposed P2P frameworks and the solutions for the VCG's inability to maintain balanced budgets, the authors conducted several case studies. An energy trading market concept that is fully P2P is presented in Alskaf et al. (2021), the study incorporates two methods of identifying market preferences. In the first strategy, excess supply and demand are matched, whereas in the subsequent method, the distance between participants is considered. Moreover, the two strategies are evaluated and compared in terms of the effect they have on the price and amount of traded locally energy. The study explains how P2P energy trading market can be developed onto the blockchain in a decentralized manner. The authors emphasized on P2P energy trading model for private houses, integrating electric heating model and optimization model.

Electricity can be purchased and sold locally by plug-in hybrid electric vehicles (PHEVs) in smart grids by using a peer-to-peer (P2P) approach. The authors provide incentives to PHEVs so that local electricity demand balances with local fleets of PHEVs instead of traditional methods that distribute electricity long distances and through complex electricity networks. As part of the study, the authors examine a promising consortium blockchain technology to enhance transaction security without requiring any mediator. The proposed system illustrates the detailed operations of localized P2P electricity trading with the consortium blockchain method (PETCON). To maximize social welfare, an iterative double auction mechanism is used to determine the price of electricity and the amount of electricity traded among PHEVs. Using PETCON, authors increase security and privacy of transactions. The application of the double auction mechanism to a real map of Texas shows that social welfare can be maximized while protecting PHEV privacy (Kang et al., 2017). In another research article, an energy trade system that employs P2P energy trading while simultaneously considering demand response is presented. The energy market is conceptualized as a non-cooperative Stackelberg game amongst prosumers and auctioneers. Prosumers, as buyers, maximize social welfare by solving optimization problems. The auctioneer also boosts the average social welfare of sellers. The authors proposed a unique Stackelberg Equilibrium (SE) that is the ideal solution to the optimization problem. Blockchain technology is used by the authors to implement the proposed algorithm, which provides enhanced security and privacy protection, and demonstrates the feasibility of the P2P energy trading approach. This blockchain-based energy trading approach is also measured for throughput, latency, and running time (Doan et al., 2021).

A latest research presents simulations results of P2P energy trading based on real-world data by introducing a market mechanism and a decentralized market clearing method with the blockchain integration. Markets and blockchain are two key components of the proposed platform. The market layer is based on

a short-term pool-supported auction and is cleared through a unique decentralized Ant-Colony Optimization procedure. Using this market procedure, players' privacy is preserved, and they can trade products between different time periods. Blockchain technologies facilitate real-time settlements through automated processes, security, and the implementation of smart contracts. In this study, the authors simulated energy trading, clearing markets, smart contract operation, and blockchain-based settlement conducted through a platform based on real-world data (Esmat et al., 2021). Ref. Vieira and Zhang (2021) examine the possibilities for self-governing peer-to-peer energy trading from within microgrids utilizing blockchain technologies and auction mechanisms. A continuous double auction framework and a uniform-priced double-sided auction framework are designed that both employ Ethereum's smart contract functionality. A/B tests were performed on a real-time data to validate the proposed design. To compare the two frameworks, the authors presented a number of cost analyses. The authors emphasized a P2P trading platform that incorporates blockchain technologies and agent-based systems could revamp the current centralized energy grid.

During peak hours, a central power system can decrease energy demand of its customers by using a peer-to-peer (P2P) energy trading approach. As a result, a cooperative Stackelberg game is devised, in which the central power system acts as the leader that determines a price during the peak demand period to compensate for prosumers not consuming energy from it. By contrast, prosumers act as followers and form coalitions with neighbors to take part in peer-to-peer energy trading to meet their energy needs in reaction to the leader's decision. Stackelberg's proposed game is examined for its properties. Because of the stability of prosumers' coalitions, the game uses a unique and stable Stackelberg equilibrium. At equilibrium, the leader formulates its strategy using a closed-form expression, while the prosumers select their coalition structure. Peer-to-peer (P2P) energy trading is suggested in this paper to lower the total electricity demand of a central power system during peak hours. In this study, the authors propose an algorithm that facilitates the equilibrium solution between the centralized and the prosumer power systems. The proposed scheme has been proven beneficial in numerous case studies (Tushar et al., 2020b). With the help of Ethereum's blockchain and smart contracts, the authors enabled peer-to-peer (P2P) energy trading amongst consumers and producers. Smart contracts reside on a blockchain shared by the participants, thus guaranteeing exact execution of trades, and keeping immutable records of transactions. The system eliminates the high costs and overheads of conventional server-based P2P energy trading. The study implements, Microgrid dynamic pricing to balance total supply and total demand, preventing double sales, automatic and autonomous operation (Song et al., 2021).

A wealth of literature review has been carried out as a part of this research and some is presented here in this paper. Our research has not found any publications referring to the usage of a locally installed Node-Red IoT server for monitoring and controlling energy in a Peer-to-Peer energy trading system design. Moreover, the use of ESP32-S2 micro-controller in our design, ensures that a local participants of P2P energy trading network can visualize the state of the process by visualizing data information on the Node-Red dashboard. A private Ethereum's blockchain is also integrated with a web interface developed using ReactJS library for performing trading activities on a locally installed server. This not only helps in performing trading activities but also to maintain the tamper-proof records of energy trading sessions. Both the servers are fully automated and interconnected with each other by means of http request method. The proposed platform is locally installed on a machine and can be accessed remotely.

This article presents a novel peer-to-peer trading platform, with the following key contributions:

- This paper presents an innovative Peer-to-Peer energy trading platform design using open-source resources with real time settlements, technical and economic efficiency.
- Design and installation of server on a local machine and communication channel makes it useful for remote locations with no internet access.
- Implementation of a Peer-to-Peer energy trading platform by installing a decentralized web interface, Ethereum blockchain, locally installed Node-Red IoT server hosted on a local network is the main feature. The fact that an open source, peer-to-peer energy trading network has never been seen before is novel to the best of our understanding.
- After a thorough review of the available research, we present a straightforward and unique approach to the design and implementation of an IoT and blockchain-based open source P2P energy trading platform. Any future participant planning to carry out such an exercise will find this paper a useful guide.

The remaining parts of the paper are arranged as follows. Section 2 of the paper presents an overview of the technologies used in this paper including blockchain, internet of things and MQTT protocol. Section 3 describes a brief overview of the proposed open-source IoT and blockchain-based Peer-to-Peer (P2P) energy trading platform. Section 4 represents the details of the components used for the design and implementation of the proposed P2P energy trading platform. This includes ESP32-S2 microcontroller, relay, current sensor, voltage sensor, Energy storage system (ESS)/battery and load. In Section 5 we discussed the web interface and IoT server and http request method to connect the servers. Node-Red flow, based on Node-Red visual programming language are also discussed in this section. Section 6 depicts the implementation methodology used in this paper. It describes how the required data is collected and energy trading sessions are performed. As a part of Section 7 system design, implementation and experimental setup are discussed. Section 8 includes the testing and results of the proposed P2P energy trading platform. In Section 9 discussion about the key features of the system is presented. In Section 10, we conclude the paper and Section 11 depicts the future work direction.

2. A brief overview of technologies

Throughout Section 2, the major technologies incorporated into the design and implementation of our proposed open-source peer-to-peer energy trading platform are briefly introduced. Ethereum blockchain and internet of things (IoT) are two of the technologies upon which P2P energy trading platform is based, as well as Message Queuing Telemetry Transport (MQTT), a lightweight protocol for IoT data transfer.

2.1. Blockchain technology

A blockchain is made up of blocks put together in the form of a chain. Data from each of the participants in the network is stored in these blocks (Al-Fuqaha et al., 2015). This is done in a block format, called a set of transactions. Following their formation, these transactions are grouped into blocks and connected to one another in chains — along with the time stamp (Baig et al., 2020). These blocks are records of data that are continuously added into a distributed ledger called a blockchain (Tapscott and Tapscott, 2016). An additional layer of security is provided by this chain mechanism, even the smallest modifications invalidate subsequent blocks. Furthermore, if only the

hash of the last block is verified, the validity of entire chain can be ascertained. Initial applications of blockchain were used to track economic transactions without requiring trusted intermediaries (such as banks). Even if its origins are in economic transactions, like Bitcoin, the blockchain has proven to be an applicable technology to other fields, when distributed approaches are the choice to centralized ones. For energy purchases or sales, blocks can be organized into tables with details such as seller ID, buyer ID, amount of transferred energy, duration, timestamp, and power profile (Di Silvestre et al., 2018). As part of this study, we have used a local Ethereum blockchain, which is provided by Ganache graphical user interface (GUI). Ganache GUI is a personal Ethereum blockchain which can be used for distributed application development. It ensures that dApps can be developed, deployed, and tested in a deterministic and secure environment (Ganache GUI documentation, 2021). Users can deploy smart contract and run tests using Ganache. Ganache also provides 10 accounts with 100 fake Ethers in each account. These Ethers and accounts can be used to perform test runs. To manage the wallet and to perform transaction MetaMask is used which is a google chrome extension. Peers willing to use P2P energy trading platform can be connected to each other on a blockchain server using remote procedure call (RPC) protocol. Fig. 1 shows the overview of Ganache GUI, a local Ethereum blockchain used as a part of this research. The digital address of each account can be seen in Fig. 1 with 100 ETHS in each account. Each account is also associated with a unique private key that ensures the security of wallet.

2.2. Internet of Things (IoT)

In this high-tech age, the Internet of Things (IoT) has changed many traditional ways of living. IoT has allowed cities, homes, pollution control, energy conservation, smart transportation, and industries to undergo amazing transformations. We can sense the Internet of Things everywhere we go, constantly increasing our quality of life. Overall, IoT encompasses a wide variety of smart systems, structures, devices, and sensors (Kumar et al., 2019). Internet of Things (IoT) combines sensor, computing, embedded, and communication technologies. The Internet of Things aims to offer smooth services to anyone, anywhere, at any time. IoT is bringing the fourth major revolution following the internet and Information and Communication Technologies (ICT). Researchers and developers predict IoT will improve society's well-being and industries more than the internet and ICT (Swamy and Kota, 2020). Basically, the internet of things (IoT) lets electronic, sensor, and software devices to interconnect with each other, as well as an operator, to facilitate the collection and sharing of real time data (Al-Fuqaha et al., 2015; Sultana and Wahid, 2019). It allows physical devices to be made smarter through the transformation of traditional forms into smart ones (Al-Fuqaha et al., 2015).

Smart energy management systems, smart emergency response systems, smart healthcare systems, smart transportation and industrial automation, smart homes and smart cities, etc. have been developed in the recent years, utilizing the IoT model (Al-Fuqaha et al., 2015; Sultana and Wahid, 2019). Also shown in Fig. 2, various smart platforms are connected with an IoT platform.

Node-Red has been selected for use as a preferred IoT platform. Locally installed Node-Red server on a private machine ensures platform's security and privacy. Node-RED allows users to replace common coding tasks with visual drag-and-drop to connect web services and gadgets. Various components are linked together to create a flow using Node-Red editor.

ACCOUNTS	BLOCKS	TRANSACTIONS	CONTRACTS	EVENTS	LOGS	SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 0	GAS PRICE 20000000000	GAS LIMIT 6721975	HARDFORK MUIRGLACIER	NETWORK ID 5777	RPC SERVER HTTP://127.0.0.1:7545	MINING STATUS AUTOMINING
WORKSPACE QUICKSTART	SAVE	SWITCH	SETTINGS			
MNEMONIC ? recipe spirit core advice menu wire outside master better casual advice route						HD PATH m/44'/60'/0'/0/account_index
ADDRESS 0x46D267500D5FD5dFD822C7D87AFce0368Ead5B62	BALANCE 100.00 ETH	TX COUNT 0	INDEX 0			
ADDRESS 0xb20E98795e82B17CC771b09174e7aAD5F6075733	BALANCE 100.00 ETH	TX COUNT 0	INDEX 1			
ADDRESS 0xb83aA8f95FbFadF6d91AE2fAbdbd8dF6d2884ab4	BALANCE 100.00 ETH	TX COUNT 0	INDEX 2			
ADDRESS 0x60E1D98a6F612D640b6f31AceE95015cf57eF2D4	BALANCE 100.00 ETH	TX COUNT 0	INDEX 3			
ADDRESS 0xAB002883f9829da4bBDB4C5dfE6d0E1F8001315c	BALANCE 100.00 ETH	TX COUNT 0	INDEX 4			
ADDRESS 0xd8D171eaA3fBeB93adCeD12dA4A51f35cDcC927D	BALANCE 100.00 ETH	TX COUNT 0	INDEX 5			

Fig. 1. Blockchain server.

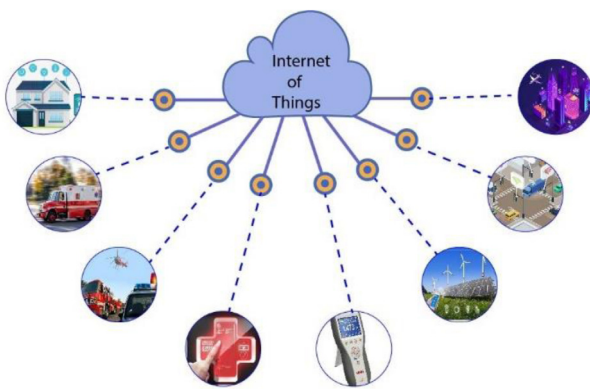


Fig. 2. Internet of things platform.

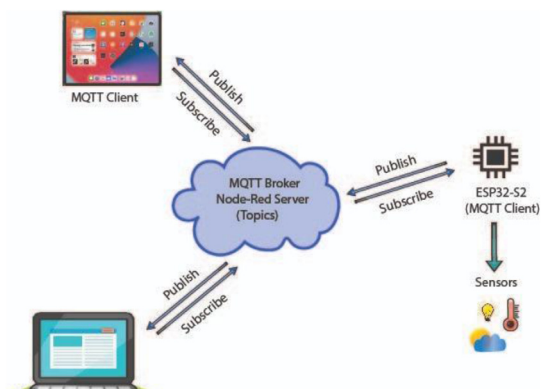


Fig. 3. Overview of MQTT protocol.

2.3. Message Queuing Telemetry Transport (MQTT) protocol

A lightweight data communication for machine-to-machine interaction can be achieved using Message Queuing Telemetry Transport (MQTT) protocol (Al-Fuqaha et al., 2015). MQTT's 2-byte fixed header makes it ideal for internet of things applications, since it can handle applications having low bandwidths, low computing power, little memory, and low batteries, which are common in IoT applications (Kodali and Soratkal, 2016; Sahadevan et al., 2017). Andy Stanford-Clark of IBM and Arlen Nipper of Arcom (now Eurotech) initially developed MQTT protocol and now the Organization for the Advancement of Structured Information Standards (OASIS) recognize MQTT as an open standard (Dhar and Gupta, 2016). For the purpose of implementing MQTT, TCP/IP connection is required, with wired or wireless local area networks (Nuratch, 2018; Lawrence O. Aghenta and Tariq Iqbal, 2020). As part of the MQTT protocol, message delivery is graded according to Quality of Service (QoS) levels 0 through 2. As a result, the MQTT protocol is ideal for IoT applications with limited resources as a reliable data transfer protocol (Al-Fuqaha et al., 2015; Kodali and Soratkal, 2016; Sahadevan et al., 2017; Lawrence O. Aghenta and Tariq Iqbal, 2020). MQTT is therefore considered preferable to other protocols like REST, CoAP, API and

HTTP, etc. MQTT also uses TCP/IP to communicate, only CoAP uses UDP (Al-Fuqaha et al., 2015; Lawrence O. Aghenta and Tariq Iqbal, 2020; Sultana and Wahid, 2019). There is an extensive discussion of the properties, benefits, and drawbacks of various IoT protocols, like MQTT, HTTP and CoAP in Al-Fuqaha et al. (2015) and Sultana and Wahid (2019). For the proposed open-source IoT and blockchain-based Peer-to-Peer Energy trading platform we have used MQTT, IPv4, TCP and IEEE 802.11 IoT protocol. MQTT protocol is used to transfer data from ESP32-S2 (MQTT client) to Node-Red IoT server. Whereas, personal smart devices (MQTT clients) can be used to subscribe topics to visualize data on Node-Red IoT server. An overview of the MQTT protocol is shown in Fig. 3.

3. The system description

Throughout this section, the authors describe the architecture of the proposed open-source P2P energy trading platform. The system configuration in Fig. 4. shows a web interface hosted on a local machine to perform energy trading tasks. The web interface is connecting to the Node-Red server used to monitor and control energy. Http request method is implemented to connect the two servers, so that an energy trading process can take

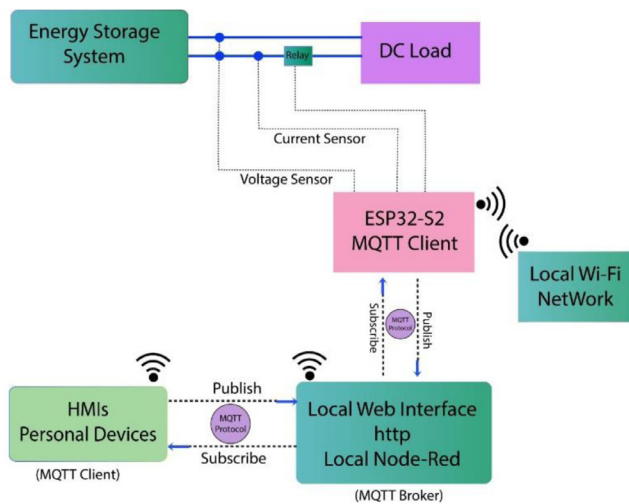


Fig. 4. P2P energy trading platform architecture.

place. To develop the communication link between the various components of the proposed system a Wi-Fi router is used. For system security, user authentication is set up on the router. Only authorized user can access the network.

Energy storage system (ESS) represents a battery which is connected to a DC load. ESP32-S2 (MQTT client) publishes the sensor data (voltage and current) to Node-Red IoT server (MQTT Broker) after processing. This configuration also demonstrates a relay device that facilitates energy trading sessions. On receiving an energy demand, it starts energy transfer process and after the demand is successfully fulfilled it halts the process. This is done by means of an algorithm without any human intervention. The participating peers can access the system using personal devices on the same network.

4. Components of proposed peer to peer energy trading platform

4.1. Sensors

4.1.1. ACS 712 hall effect current sensor

ACS 712 Hall effect current sensor is used to measure load current using Hall Effect principle. It provides voltage isolation of 2.1 kVRMS and a low resistance conductor that is integrated into the sensor. Hall ICs convert the magnetic field generated by copper conductors into proportional output voltages by converting the current flowing through the copper conductors. As a result, an output voltage is produced which is proportional to the measured AC or DC current (Current Sensors ICs, 2021). This module is available to sense the current of 05 A, 20 A and 30 A with the output sensitivity range of 66 mV/A to 185 mV/A at a supply voltage of 5VDC. It uses the ACS712ELCTR chip as its central component (Current Sensors ICs, 2021). To measure analog signals, the ESP32-S2 uses a maximum range 3.3VDC supply, so by connecting the Vcc pin ACS-712 Hall effect current sensor with the 3.3 V pin of this sensor and selecting appropriate value of sensitivity accuracy of measured values is ensured. As a part of this study, we have used 20 A DC module.

4.1.2. Voltage sensor

ESP32 is a 13-bit device, and its ADC pins are capable to read 3.3 V. To convert the bus voltage of the proposed open-source P2P energy trading platform to the readable signal range of ESP32-S2

ADC pins, a simple voltage divider circuit is used in this project. Eq. (1) shows the voltage divider equation.

$$V_{out} = V_{in} \times \frac{R_2}{(R_1 + R_2)} \quad (1)$$

where V_{out} = ESP32-S2 ADC voltage (3.3) and V_{in} = Source voltage or Battery voltage (12 V), Whereas R_1 and R_2 are calculated using V_{out} and V_{in} as a reference.

With voltage divider circuit, the bus voltage of 12 V is converted to 3.3 V, which ensures the proposed design is capable to sense the bus voltage of the proposed platform.

4.2. ESP32-S2 micro-controller

ESP32-S2 a single core Wi-Fi microcontroller is designed by ESPRESSIF and featured with high performance, low power consumption, and a vast array of I/O (inputs and outputs). An integrated product featuring extremely low power consumption and 2.4 GHz Wi-Fi on a system-on-chip solution makes it ideal for IoT applications. The Wi-Fi solution is offered by this chip fully complies with the IEEE 802.11b/g/n protocol. There is a 32-bit Xtensa® LX7 processor on this chip that operates at 240 MHz. This device has a rich set of peripheral interfaces including I2S, I2C, UART, SPI interface, LED PWM, LCD, camera interface, ADC, touch sensor, DAC and temperature sensor, in addition to 43 GPIOs (General Purpose Inputs and Outputs). In addition, it supports full-speed USB On-The-Go (OTG), which allows USB communications (ESP32-S2-WROOM product details, 2021).

4.3. Wi-Fi router (communication channel)

For data communication on the TCP/IP Wireless Network between ESP32-S2 (MQTT Client) and Node-Red server (MQTT Broker) and web interface we have used D-Link Router (DI-524 Airplus G). To establish a local Wi-Fi connection network, the router is used, since the ESP32-S2 microcontroller used in this project is compatible with TCP/IP and IEEE 802.11b/g/n Wi-Fi standards (Lawrence O. Aghenta and Tariq Iqbal, 2020; ESP32-S2-WROOM product details, 2021). In this way ESP32-S2 publishes sensor data to Node-Red IoT server over MQTT protocol. Wi-Fi network connections is restricted to authorized persons using service set identifier (SSID) and password.

For the successful testing and to obtain required results, we have also used a 12 V, 7.2 Ah/20HR battery as source voltage and 12 V, 50 W DC light bulb as load. We have also developed and interactive user interface to perform energy transactions using React.JS and a Node-Red server is set up to monitor and control the self-generated energy. More details about the server are presented in the following section of the paper. Table 1. presents the name of the components used to design an open-source P2P energy trading platform.

5. Web interface and node-red IoT server

We have used React.JS (an open-source tool used to create user interfaces for web applications) library to create front-end/web interface for the proposed open-source P2P energy trading platform which is also locally installed on the system and accessible from <http://localhost:3000/> and is integrated with MetaMask and Ganache GUI. The smart contract has been deployed on the local Ethereum blockchain. Reloading is made faster with the virtual document object model (DOM) since the virtual DOM allows rendering to be sped up. The React.JS framework is also used by many real-time applications, including Facebook and Netflix (Saundariya et al., 2021). The source code used for web interface, connectivity with Ganache GUI and smart contract are extensions of Ethereum Marketplace Step-by-Step Tutorial (2021) to

Table 1
Components used in P2P energy trading platform.

Module used	Purpose of module in the design
ReactJS library	To develop web interface of the system
Node-Red server	To monitor and control self-generated energy
Ganache GUI	To deploy local Ethereum blockchain
ESP32-S2 Microcontroller	Read sensors value and publish sensors data over MQTT
ACS-712 hall effect current sensor	To measure load current
Voltage sensor	To measure the voltage
Relay	To start or stop energy transfer
12 V, 7.2 Ah/20HR battery	Source voltage
12 V, 50 W DC light bulb	To serve as load
Electric wires	To connect the components of proposed design
Jumper wires	To connect the components of proposed design
Breadboard	Build the circuit

develop an appropriate P2P energy trading platform. The detailed installation guide on the basic concept can be found at [How to Build \(2021\)](#). Participating peers of the proposed open-source P2P energy trading platform can access the user interface at <http://localhost:3000/> to perform energy trading tasks. Fig. 5. shows an interactive and user-friendly interface developed for the P2P energy trading with a MetaMask plugin. The user interface is linked with local Ethereum blockchain on which smart contract is deployed. All the transaction performed are saved on an immutable blockchain network. As shown in Fig. 5. participating peers can bid their energy requirement and in response the peers with access energy can also post a selling notification with price and quantity of energy. This allows the buyer to buy energy on a most suitable price, which may be less than the bidding price of the buyer. The user interface developed for P2P energy trading platform is integrated with Ganache GUI a private Ethereum blockchain which can be accessed through MetaMask. Peers can access their respective wallets using MestaMask plugin. Once the peers perform any type of transaction on the user interface MetaMask plugin will pop up automatically on the screen and peers will reconfirm the amount charged for their particular transaction, that might be the energy purchasing price or gas used. Gas is a fee Ethereum charges to successfully perform the transactions on a blockchain network. Ethereum blockchain network charges a fraction of Ether in terms of fee. The price of gas is expressed in gwei, a unit of ETH, and each gwei equals 0.000000001 ETH (10⁻⁹ ETH). Rather than saying that your gas costs 0.000000001 ether, you might say it costs 1 gwei ([Ethereum Gas Fees, 2021](#)). The MetaMask notification is shown in Fig. 6 is linked with <http://localhost:3000/> and allows to access accounts on a local Ethereum blockchain. For the security of the system the peers on the P2P energy trading platform use their own password to access the platform as shown in Fig. 6, in addition each account is associated with a unique private key, which allows the user to access their Ethereum wallets.

The proposed system incorporates the features of a fully decentralized network, the participating peers have full liberty to decide the energy quantity and price without the involvement of any central authority. Peers can also delete the posted bid/offer if for some reason the transaction is not complete.

Node-RED an open-source solution was developed by IBM Emerging Technology. This web application is based upon Node.js (a server-side java scripting platform). Node-RED runs locally on <http://localhost:1880>. It also offers options to quickly build a live data dashboard using Node-RED user interface module. The user interface is accessible from <http://localhost:1880/ui> using a local browser ([Kodali and Anjum, 2018](#)). Node-Red flow can be made

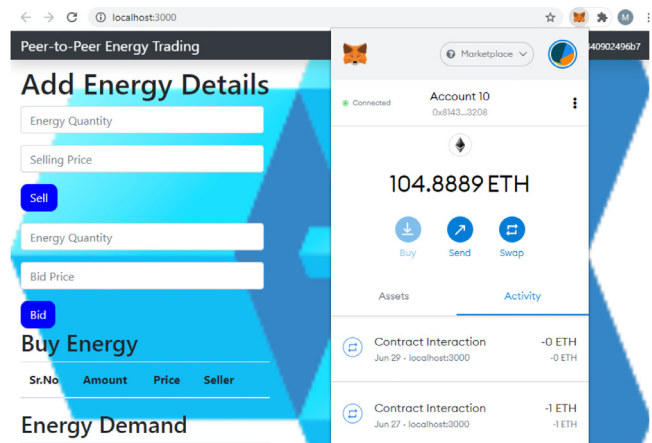


Fig. 5. Web interface of proposed P2P platform with MetaMask add on.

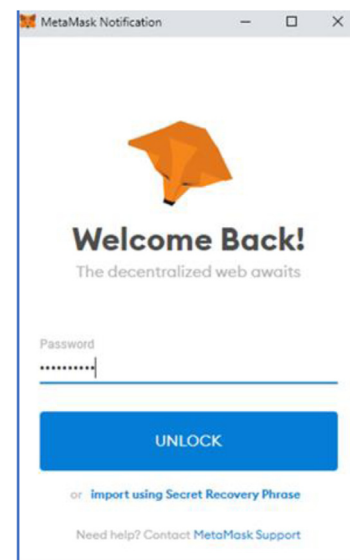


Fig. 6. MetaMask plugin.

by using simple drag and drop options and flow can also be saved, import and export as a JSON file.

Fig. 7 shows a Node-Red flow programmed using Node-Red visual programming language. The voltage, current, power and battery state f charge (SoC) data are transferred via MQTT protocol from the hardware components of the system and can be visualized on Node-Red user interface. Node-Red server is also installed on a local machine.

For the fully automation of the P2P energy trading platform, we have implemented Hypertext Transfer Protocol (HTTP) request method. There are different http request methods like, DELETE, POST, PUT, GET, HEAD, OPTIONS, PATCH. We have used http POST method to catch the request from <http://localhost:3000/>. Once the participant peer of P2P energy trading platform sends a buy request to purchase energy, the amount of energy will be posted to Node-Red server and if the required amount of energy is greater than zero, it will be transferred automatically using a computer algorithm. The algorithm is implemented in the function node on Node-Red flow that will compare the energy demand with the amount of energy being transferred. Once the required amount of energy is transferred it will send a turn off message to the relay. In this way we have attained a fully automated P2P energy trading model on a local blockchain using open-source technology.

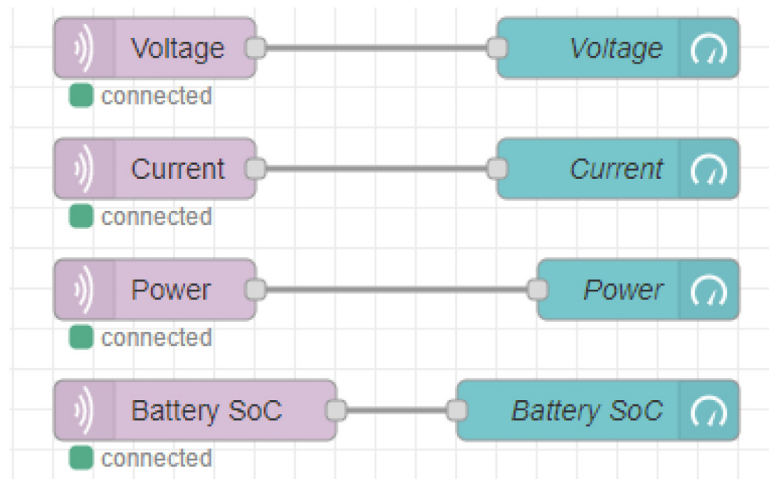


Fig. 7. Node-Red flow for data acquisition.

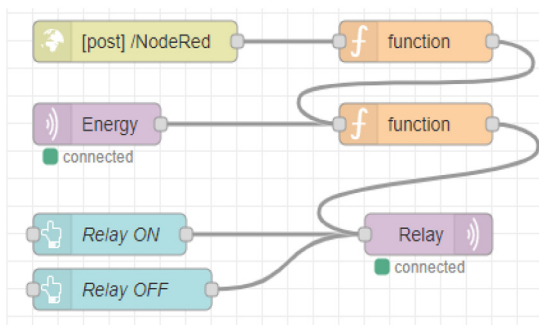


Fig. 8. Node-Red flow to facilitate energy trading.

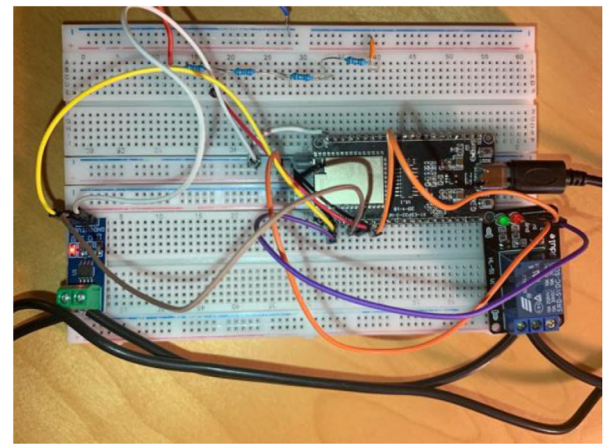


Fig. 9. Hardware design for the proposed P2P energy trading platform.

Fig. 8. shows a Node-Red flow that facilitate energy trading with a http request node to get the value of required energy from the web interface of the proposed P2P energy trading platform. Once the Node-Red https request node gets the energy demand from the web interface and then the attached function node converts the JSON string into the raw value to compare with the MQTT energy node using the other function node shown in Fig. 8 attached directly with MQTT energy node. When the energy demand and energy transferred amounts are equal, compared by implementing the algorithm in the function node it will actuate the relay to stop energy transfer process.

6. Implementation methodology

As a part of implementing the proposed open-source P2P energy trading platform current, voltage, and battery state of charge (SoC) data are measured and processed. Electrical wires are used to connect the current and voltage sensors to the system. The ESP32-S2 allows the measurement and collection of data using the Arduino IDE programs. ESP32-S2 micro-controller has been programmed using Arduino IDE programs.

The PubSubClient MQTT library is used to configure ESP32-S2 micro-controller as an MQTT client and the measured and collected data are transferred to locally installed Node-Red server (MQTT Broker) via MQTT protocol over TCP/IP Wi-Fi communication link. Node-Red server receives data and displays it on the Node-Red dashboard. Algorithm 1 shows the pseudocode that encapsulates the data flow process. In Algorithm 1 pseudocode (line 1 to line 8) explains ESP32-S2 pins read the sensors values and calculate the values of battery state of charge (SoC), power, and

energy. It also shows the connection process of ESP32-S2 with Wi-Fi and the MQTT broker. The process regarding the publishing of data to the MQTT broker is also shown in this part of the pseudocode. Line 9 to 13 presents the establishment of an ESP32-S2 connection with Wi-Fi and displays the connection status on the Arduino serial monitor with a data sampling time of 100 milli seconds. In this way data and information are processed and made available to facilitate energy trading process. In Algorithm 2 pseudocode for energy trading process is explained. Once the peer shows the desire to buy energy and starts the process using the web interface of the proposed P2P energy trading platform. The information about amount of energy required is collected at Node-Red server and energy trading process will start. The algorithm then compares the amount of energy being transferred and required energy. Once the energy demand is fulfilled the relay will automatically disconnect the system and energy transfer process will stop.

7. Hardware design and implementation

In this section of the paper the authors have described the hardware design and implementation of the proposed P2P energy trading platform. We have used a voltage sensor to measure the real time storage battery voltage. This is done by means of voltage divider rule explained in Section 4.1.2 of this paper. The current sensor shown in Fig. 9 is used to measure current drawn by

Algorithm 1: Data acquisition

```

Initialization;
1. Voltage and current data are measured using analog sensors;
2. ESP32-S2 defined pins reads sensors values and calculates values of Battery (SoC), Power and Energy;
3. ESP32-S2 connects to local Wi-Fi network using SSID and Password;
4. ESP32-S2 identifies Node-Red server/ MQTT broker using server IP Address;
5. ESP32-S2 publishes sensors data to MQTT broker using TCP/IP Wi-Fi connection;
6. ESP32-S2 publishes Power and Energy data to MQTT broker using TCP/IP Wi-Fi connection;
7. Node-Red server receives data on Node-Red flow;
8. Node-Red server displays data on Node-Red dashboard;
while
9. ESP32-S2 is not connected;
10. Establish ESP32-S2 connection by passing, SSID, and password;
if
11. ESP32-S2 connection successfully established;
12. Display "Connected" on Arduino IDE Serial Monitor;
else
13. Display "Connection failed.... retry in 5 seconds" on Arduino IDE Serial Monitor;
end

```

Algorithm 2: Energy Transfer

```

1. Collects energy demand data from the web interface;
2. Collects energy transfer data from the Node-Red;
3. Compares energy demand data with energy being transferred;
if
4. Energy demand is greater than zero, turn on the relay and start energy transfer;
else
5. Relay will remain close and no energy transfer;
end

```

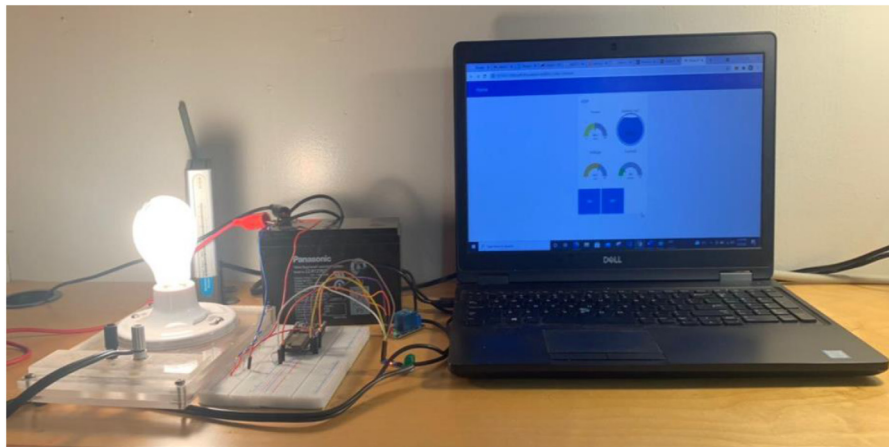


Fig. 10. Experimental setup for the proposed P2P energy trading platform.

the load and the relay shown will start/stop the trading sessions automatically. ESP32-S2 micro-controller depicted in Fig. 9 is used to collect and process data and to control relay operation.

8. Experimental setup for the proposed peer-to-peer energy trading platform

As described in Section 7 the design of proposed open-source P2P energy trading platform is set up for operation. The complete experimental setup of the system is shown in Fig. 10. This set up also uses a battery and load to practically demonstrate the

energy transfer operation. The battery acting as source voltage of the system, is connected with the inputs of voltage sensors, while relay and the current sensor are set up between the battery and the load in series connection. The data can be collected by ESP32-S2 microcontroller using the sensors output pins and onwards transferred to the locally installed Node-Red IoT server. Node-Red IoT server receives energy demand form the web interface developed to initiate trading sessions and start energy transfer process automatically. The Node-Red is also connected with web interface to perform energy trading tasks (explained in Section 5 of this paper).



Sr.No	Amount	Price	Buyer	
1	1 whr	1 Eth	0x81434b8aa49223d9D6D7249A64853425Dd2b3208	Purchased
2	3 Whr	2 Eth	0x139320128e32de57391682786cB8253b5aB3A480	Purchased
3	11 whr	6 Eth	0x81434b8aa49223d9D6D7249A64853425Dd2b3208	Purchased
4	1 whr	1 Eth	0x139320128e32de57391682786cB8253b5aB3A480	Sell Delete
5	2 Whr	3 Eth	0x81434b8aa49223d9D6D7249A64853425Dd2b3208	Sell Delete
6	2 Whr	4 Eth	0x27b62Dd1eD0394018B22c2D38b13E01C83e8b862	Sell Delete

Fig. 11. Energy demands posted by the buyers.

9. Testing and results

The proposed open-source P2P energy trading platform has been tested to obtain the required results. Fig. 17 shows a flow

chart of the proposed system. The flow chart gives a brief overview of complete energy trading process.

The designed P2P energy trading setup is for 10 peers, there is no transaction delay, and the designed server can easily handle all requests. The participants may face delay in transactions on a public blockchain network like bitcoin due to a large ledger. However, in the proposed system, the ledger is small and maintained on a private server. It can process transactions immediately.

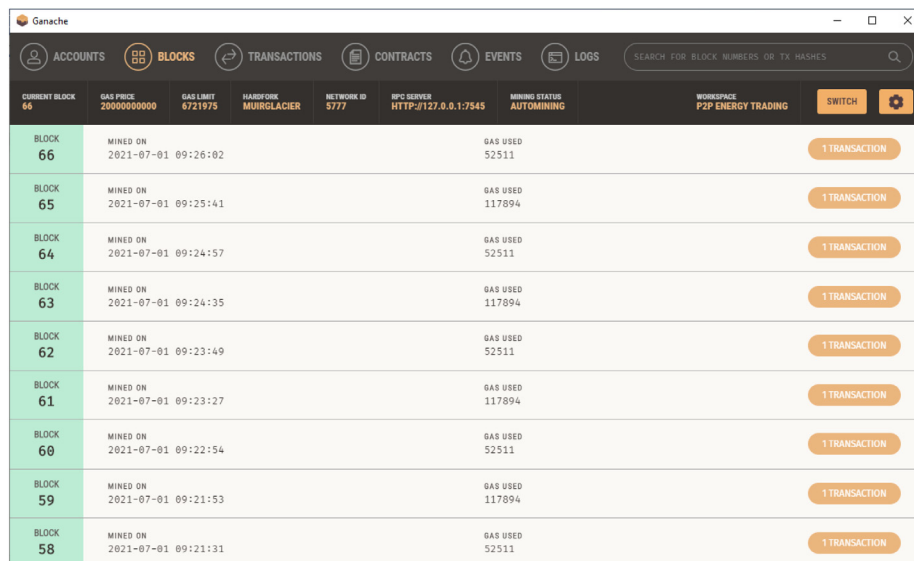
9.1. Results

The web interface incorporated with private Ethereum blockchain is developed using ReactJS library and is connected to Node-Red server, and to the hardware side of the proposed open-source P2P energy trading platform. Once a peer bids energy demand, in response the other peers on the platform can post the available energy at their premises and the price they are willing to sell the energy on the web interface details explained in Section 5 Fig. 5 of this paper. This gives the buyer an opportunity to choose amongst the various available options even with a greater amount of energy as required at a lesser price. All the energy demands posted by the peers will appear in the energy



Sr.No	Amount	Price	Seller	
9	5 Whr	1 Eth	0x81434b8aa49223d9D6D7249A64853425Dd2b3208	Purchased
10	5 Whr	1 Eth	0x139320128e32de57391682786cB8253b5aB3A480	Purchased
11	5 Whr	1 Eth	0x81434b8aa49223d9D6D7249A64853425Dd2b3208	Purchased
12	2 Whr	1 Eth	0x40234186eA4e23bb6925887D8C21b8486ce66FD5	Buy Delete
13	2 Whr	2 Eth	0x710B2322b2aA5aD0f81C7b47af023B75ecec3Fdf	Buy Delete
14	2 Whr	5 Eth	0xcaF073257dDc8F00280710f0365D31CEE26345eC	Buy Delete

Fig. 12. Available Energy posted by sellers.



BLOCK	MINED ON	GAS USED	
66	2021-07-01 09:26:02	52511	1 TRANSACTION
65	2021-07-01 09:25:41	117894	1 TRANSACTION
64	2021-07-01 09:24:57	52511	1 TRANSACTION
63	2021-07-01 09:24:35	117894	1 TRANSACTION
62	2021-07-01 09:23:49	52511	1 TRANSACTION
61	2021-07-01 09:23:27	117894	1 TRANSACTION
60	2021-07-01 09:22:54	52511	1 TRANSACTION
59	2021-07-01 09:21:53	117894	1 TRANSACTION
58	2021-07-01 09:21:31	52511	1 TRANSACTION

Fig. 13. Blockchain status on server.

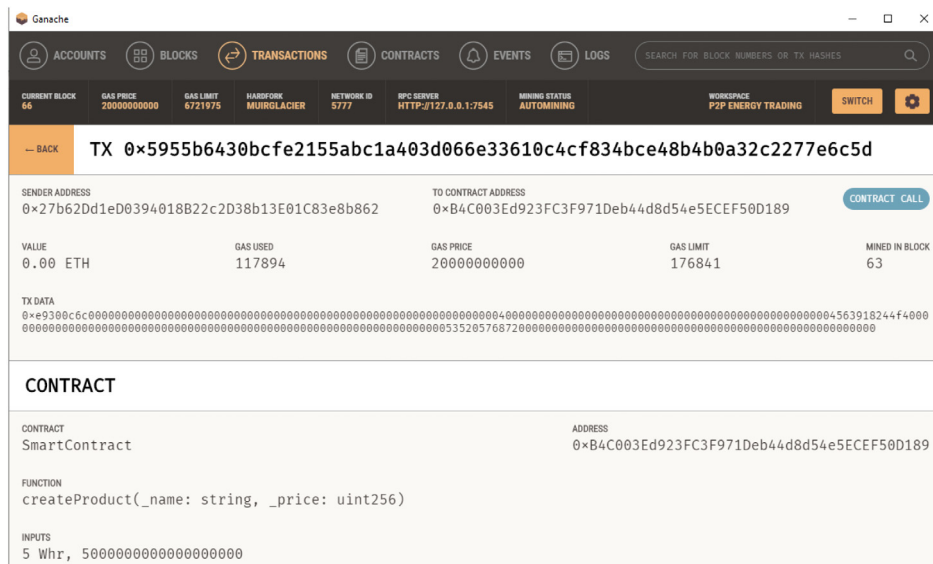


Fig. 14. Details of block 64 on the blockchain server.

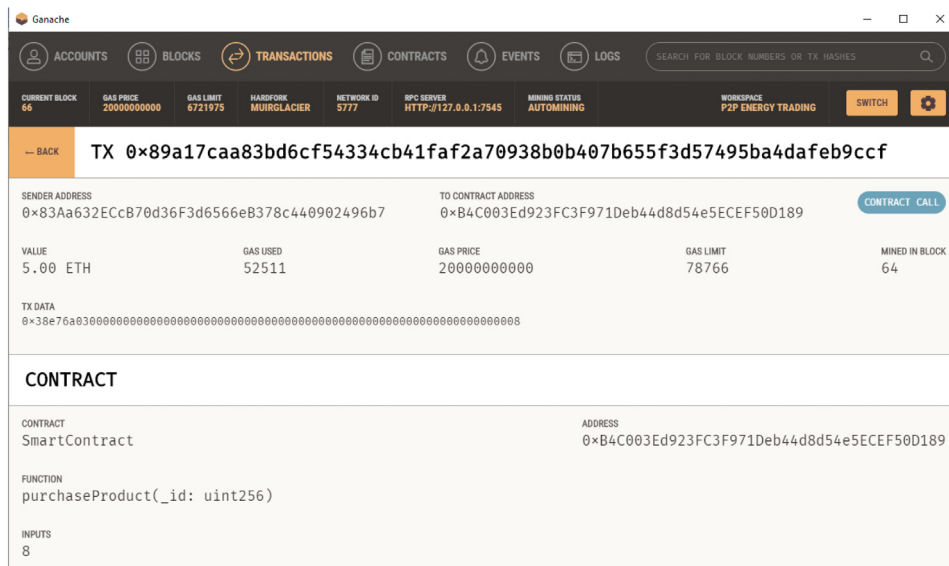


Fig. 15. Details of block 63 on the blockchain server.

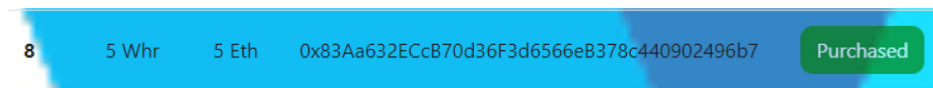


Fig. 16. Successful transaction status on web interface.

demand section as shown in Fig. 11. The other peers on the P2P energy trading platform can visualize the bids on HMLs (dashboards). In response they can post the available energy on the web interface at the price of their own choice. The amount of energy available to sell will show up in the buy energy section of the web interface. If a participating peer of the energy trading platform changes his mind after posting any trading action, there is also an option to delete.

In Fig. 11 serial no. 4, a peer is willing to buy 1 Whr of energy at a price of 1 ETH. In response of his bid, he has an option to buy say 2 Whr of energy for 1 ETH, as shown in Fig. 12 serial no.12. Similarly, another participating peer (serial no 5 Fig. 11) of the proposed P2P energy trading platform willing to buy 2 Wh

of energy at a price of 3 ETH. Whereas, he has the option to buy the required amount of energy at lesser price as shown in Fig. 12 serial no. 13. This platform gives an option to both buyers and seller to sell energy at their desired price according to the market response.

All the trading actions performed by the peers on the web interface will appear on the local Ethereum blockchain. We have performed 66 transactions using the web interface and all the transactions are recorded on the blockchain server. The blocks are mined with the timestamps and can be explored to get the details of the transaction performed. Fig. 13 depicts, 66 blocks are formed in response of different trading tasks performed using web interface.

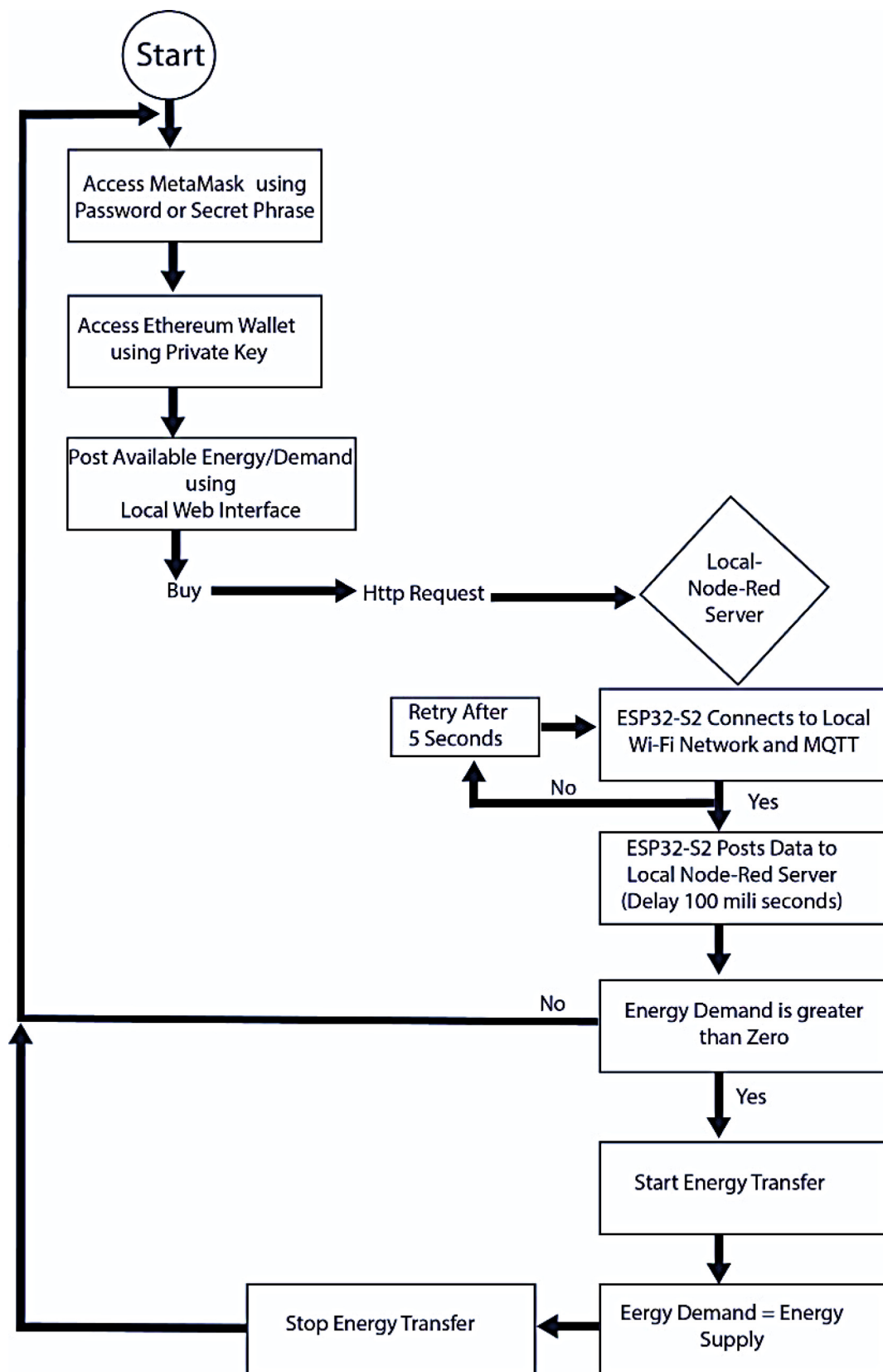


Fig. 17. Flow chart of proposed P2P energy trading platform.

Fig. 14 shows details of block 63. 0xB4C003Ed923FC3F971Deb44d8d54e5ECEf50D189 is the address where the smart contract is deployed. The energy demand posted by the seller (0x27b62Dd

1eD0394018B22c2D38b13E01C83e8b862) using the web interface and this will appear on the blockchain. A small amount of gas is used a fee to deploy the transaction on the blockchain server.

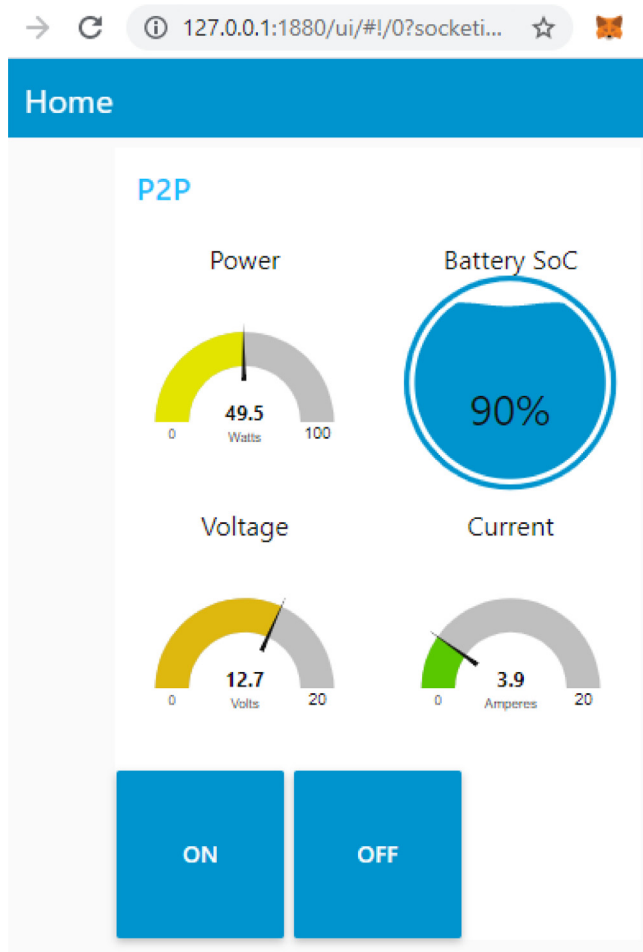


Fig. 18. Node-Red dashboard.

A unique id is assigned to this transaction called transaction hash and can be seen in block 63 as TX 0x5955b6430bcfe2155abc1a403d066e33610c4cf834bce48b4b0a32c2277e6c5d. Function in each block shows a product is created to sell in the energy market in response of the action performed at web interface.

In Fig. 15 the inside details of block 64 are shown. When we explore block 64 of the blockchain, where a buyer purchases the required amount of energy with a buyer ID 0x83Aa632ECcB70d36F3d6566eB378c440902496b7 the function now shows purchase product and an amount of 5 ETH has been charged with a some gas as a fee. The block 64 mined in response of energy purchased. It also has a unique transaction hash assigned to it. Once the trading sessions is complete it is recorded on the blockchain and can also be visualized on the web interface of the proposed P2P energy trading platform as shown in Fig. 16.

After collecting the sensor data ESP32-S2 microcontroller transmits data to Node-Red IoT server which is responsible for monitoring and control of energy. The data collected and processed by ESP32-S2 microcontroller can be visualized on Node-Red dashboard as shown in Fig. 18. The On and Off buttons shown here are developed here to initiate or stop energy trading sessions using Node-Red dashboard only.

The proposed P2P energy trading platform can be used for a small community of 10 houses within the range of local communication link setup through the Wi-Fi router, which can be increased and decreased depending on the community's need.

The Wi-Fi router used for the current project has a range of 328 Feet indoor and 1312 Feet outdoor.

Discussion

This section describes a few of the key characteristics of the proposed open-source IoT and blockchain-based Peer-to-Peer energy trading platform realized after successful testing of the system:

- **Blockchain and IoT based P2P energy trading platform** The proposed open-source P2P energy trading platform is based on IoT platform and a private Ethereum blockchain. The proposed platform introduces vital components that a blockchain-based P2P energy trading system should include for proper energy monitoring, energy metering and money transaction in a decentralized manner. This includes blockchain-based web interface, Node-Red IoT server, ESP32-S2 microcontroller, Field instrumentation devices and MQTT protocol communication channel.
- **Energy Trading** A web interface is developed for Human machine interactions. Real time settlements can be done using the web interface. This web interface can be accessed remotely to perform energy trading tasks.
- **Ethereum Blockchain:** A private Ethereum blockchain has been used to deploy smart contract and maintain digital ledger of transactions. All the trading activities performed using the web interface are recorded on an immutable blockchain network with timestamps with all the security features of blockchain.
- **Fast Transactions:** The proposed P2P energy trading platform can perform transaction on a blockchain network immediately. There is no delay in the transactions and system can handle all requests easily.
- **Data Acquisition:** The proposed system is capable of data acquisition using field instrumentations devices commonly available in the market.
- **Monitoring and Control:** Locally installed Node-Red IoT server is used to monitor and control self-generated power. Peers can monitor the available energy and are able to transfer energy in response to a buy call.
- **Security:** The proposed P2P energy trading platform is restricted to authorized users only. To access the Ethereum wallet MetaMask require login credentials plus Ethereum wallet itself has a unique private key for each user which ensures authorized access to the funds and platform. To add to this local Wi-Fi network also require user authentication credentials.
- **Central Server:** After implementing proposed P2P energy trading platform, there is no need for central servers to support the energy trading.
- **The ease of use of the system:** The proposed P2P energy trading system is developed in user friendly manner. The participant of the platform does not require any training to use the system.

10. Conclusion

In the current era where energy consumers are now becoming prosumers and a large amount of energy is generated by renewable energy resources particularly by using solar photovoltaic (PV) panels. The participants of energy market are exploring different ways to sell self-generated energy and looking for such platforms which can help them to get returns on their investment in renewable energy resources. Peer-to-Peer energy trading platforms are such platforms which makes renewable energy more accessible to everyone and participants can make better use of

self-generated energy. Such platforms can help to grow renewable energy resources and can reduce extra burden on the power system. Also, the participants of P2P energy trading platform can get profit on their investment.

In this article, the authors presented an advanced open-source P2P energy trading platform using internet of things and blockchain. We have demonstrated both the software and hardware implementations of our proposed P2P energy trading platform. The software implementation is done by installing web interface hosted on a local machine using React JS library with Ganache GUI a private Ethereum blockchain. The web interface facilitates all the trading activities which are recorded on a blockchain network. For data monitoring and energy control we have used Node-Red server that is also hosted on a local machine. In hardware implementation we have used ESP32-S2 microcontroller with field instrumentation devices. From the testing and results we proved that the proposed P2P energy trading platform works perfectly. The system operates in a decentralized manner and all the transactions are documented on digital ledger. The system performs all the trading sessions properly and monitors energy data accurately. Although, the system has been tested on a small storage 12 V battery and 50 W load, customization of the system is also possible to facilitate energy trading to fulfill the energy needs of a household.

11. Future work

As a future work the authors aim to implement this project to the remote location with poor/no electricity network and no internet access. Particularly smaller communities with abundant renewable energy sources. This project aims to facilitate such parts of the world where people are living without electricity or have electricity for shorter period during the whole day. We plan to add auto sale and purchase features along with full electrical connection details for a remote site in Pakistan. The increased range of communication link setup for this decentralized network is also the part of future work extension. Details will be published in future papers.

CRediT authorship contribution statement

Mirza Jabbar Aziz Baig: Methodology, Software and hardware implementation, Data curation, Writing – original draft. **M. Tariq Iqbal:** Conceptualization, Resources, Supervision, Funding acquisition, Writing – review & editing. **Mohsin Jamil:** Co-Supervision, Writing – review & editing. **Jahangir Khan:** Co-Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank Mirpur University of Science and Technology (MUST), Mirpur-10250, Azad Jammu and Kashmir, Pakistan, and Higher Education Commission (HEC), Islamabad-44000, Pakistan for providing an opportunity to carry out this research under Human Resource Development program.

References

- Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M., Ayyash, M., 2015. Internet of things: A survey on enabling technologies, protocols, and applications. *IEEE Commun. Surv. Tutor.* 17 (4), 2347–2376. <http://dx.doi.org/10.1109/COMST.2015.2444095>, Fourthquarter.
- AlAshery, M.K., et al., 2021. A blockchain-enabled multi-settlement quasi-ideal peer-to-peer trading framework. *IEEE Trans. Smart Grid* 12 (1), 885–896. <http://dx.doi.org/10.1109/TSG.2020.3022601>.
- Alsaikaf, T., Crespo-Vazquez, J.L., Sekuloski, M., v. Leeuwen, G., Catalao, J.P.S., 2021. Blockchain-based fully peer-to-peer energy trading strategies for residential energy systems. *IEEE Trans. Ind. Inform.* <https://doi-org.qe2a-proxy.mun.ca/10.1109/TII.2021.3077008>.
- Baig, M.J.A., Iqbal, M.T., Jamil, M., Khan, J., 2020. IoT And blockchain based peer to peer energy trading pilot platform. In: 2020 11th IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON). pp. 0402–0406. <http://dx.doi.org/10.1109/IEMCON51383.2020.9284869>.
2021. Current sensors ICs. Available online: <https://www.digikey.ca/en/products/detail/allegro-microsystems/ACS712ELCTR-20A-T/1284594> [Accessed on June 23, 2021].
- Dhar, P., Gupta, P., 2016. Intelligent parking cloud services based on IoT using MQTT protocol. In: 2016 International Conference on Automatic Control and Dynamic Optimization Techniques (ICADOT). pp. 30–34. <http://dx.doi.org/10.1109/ICADOT.2016.7877546>.
- Di Silvestre, M.L., Gallo, P., Ippolito, M.G., Sanseverino, E.R., Zizzo, G., 2018. A technical approach to the energy blockchain in microgrids. *IEEE Trans. Ind. Inf.* 14 (11), 4792–4803. <http://dx.doi.org/10.1109/TII.2018.2806357>.
- Doan, H.T., Cho, J., Kim, D., 2021. Peer-to-peer energy trading in smart grid through blockchain: A double auction-based game theoretic approach. *IEEE Access* 9, 49206–49218. <http://dx.doi.org/10.1109/ACCESS.2021.3068730>.
- Esmat, A., de Vos, M., Ghiassi-Farrokhfal, Y., Palensky, P., Epema, D., 2021. A novel decentralized platform for peer-to-peer energy trading market with blockchain technology. *Appl. Energy* 282, 116123. <http://dx.doi.org/10.1016/j.apenergy.2020.116123>.
2021. Esp32-S2-WROOM product details. Available online: https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf [Accessed on June 24, 2021].
2021. Ethereum gas fees. Available online: <https://ethereum.org/en/developers/docs/gas/> [Accessed on June 30, 2021].
2021. Ethereum marketplace step-by-step tutorial. Available online: <https://github.com/dappuniversity/marketplace> [Accessed on May 30, 2021].
2021. Ganache GUI documentation. Available online: <https://www.trufflesuite.com/docs> [Accessed on June 23, 2021].
2021. How to build a blockchain app with ethereum, web3.js & solidity smart contracts. Available online: <https://www.dappuniversity.com/articles/how-to-build-a-blockchain-app> [Accessed on May 30, 2021].
- Kang, J., Yu, R., Huang, X., Maharjan, S., Zhang, Y., Hossain, E., 2017. Enabling localized peer-to-peer electricity trading among plug-in hybrid electric vehicles using consortium blockchains. *IEEE Trans. Ind. Inf.* 13 (6), 3154–3164. <http://dx.doi.org/10.1109/TII.2017.2709784>.
- Kodali, R.K., Anjum, A., 2018. IoT based HOME automation using node-RED. In: 2018 Second International Conference on Green Computing and Internet of Things (ICGCIoT). pp. 386–390. <http://dx.doi.org/10.1109/ICGCIoT.2018.8753085>.
- Kodali, R.K., Soratkal, S., 2016. MQTT Based home automation system using ESP8266. In: 2016 IEEE Region 10 Humanitarian Technology Conference (R10-HTC). pp. 1–5. <http://dx.doi.org/10.1109/R10-HTC.2016.7906845>.
- Kumar, S., Tiwari, P., Zymbler, M., 2019. Internet of things is a revolutionary approach for future technology enhancement: a review. *J. Big Data* 6, 111. <http://dx.doi.org/10.1186/s40537-019-0268-2>.
- Lawrence O. Aghenta, Tariq Iqbal, M., 2020. Design and implementation of a low-cost, open source IoT-based SCADA system using ESP32 with OLED, ThingsBoard and MQTT protocol. *AIMS Electron. Electr. Eng.* 4 (1), 57–86. <http://dx.doi.org/10.3934/ElectrEng.2020.1.57>.
- Nuratch, S., 2018. Applying the MQTT protocol on embedded system for smart sensors/actuators and IoT applications. In: 2018 15th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON). pp. 628–631. <http://dx.doi.org/10.1109/ECTICon.2018.8619981>.
- Sahadevan, A., Mathew, D., Mookathana, J., Jose, B.A., 2017. An offline online strategy for IoT using MQTT. In: 2017 IEEE 4th International Conference on Cyber Security and Cloud Computing (CSCloud). pp. 369–373. <http://dx.doi.org/10.1109/CSCloud.2017.34>.
- Saundariya, K., Abirami, M., Senthil, K.R., Prabakaran, D., Srimathi, B., Nagarajan, G., 2021. Webapp service for booking handyman using mongodb, express JS, react JS, node JS. In: 2021 3rd International Conference on Signal Processing and Communication (ICPSC). pp. 180–183. <http://dx.doi.org/10.1109/ICSPC51351.2021.9451783>.

- Song, J.G., Seon Kang, E., Shin, H.W., Jang, J.W., 2021. A smart contract-based P2P energy trading system with dynamic pricing on ethereum blockchain. *Sensors* 21 (6), 1985, [Online]. Available: <http://dx.doi.org/10.3390/s21061985>.
- Soto, E.A., Bosman, L.B., Wollega, E., Leon-Salas, W.D., 2021. Peer-to-peer energy trading: A review of the literature. *Appl. Energy* 283, 116268. <http://dx.doi.org/10.1016/j.apenergy.2020.116268>.
- Sultana, T., Wahid, K.A., 2019. Choice of application layer protocols for next generation video surveillance using internet of video things. *IEEE Access* 7, 41607–41624. <http://dx.doi.org/10.1109/ACCESS.2019.2907525>.
- Swamy, S.N., Kota, S.R., 2020. An empirical study on system level aspects of internet of things (IoT). *IEEE Access* 8, 188082–188134. <http://dx.doi.org/10.1109/ACCESS.2020.3029847>.
- Tapscott, D., Tapscott, A., 2016. *Blockchain revolution: How the technology behind bitcoin is changing money business and the world*.
- Tushar, W., Saha, T.K., Yuen, C., Smith, D., Poor, H.V., 2020a. Peer-to-peer trading in electricity networks: An overview. *IEEE Trans. Smart Grid* 11 (4), 3185–3200. <http://dx.doi.org/10.1109/TSG.2020.2969657>.
- Tushar, W., et al., 2020b. Grid influenced peer-to-peer energy trading. *IEEE Trans. Smart Grid* 11 (2), 1407–1418. <http://dx.doi.org/10.1109/TSG.2019.2937981>.
- Vieira, G., Zhang, J., 2021. Peer-to-peer energy trading in a microgrid leveraged by smart contracts. *Renew. Sustain. Energy Rev.* 143, 110900. <http://dx.doi.org/10.1016/j.rser.2021.110900>.